

The background of the slide is a dark blue gradient. Overlaid on this are several large, semi-transparent spheres in various colors including teal, purple, light green, and orange. Some of these spheres have a small, bright yellow-orange spot in the center, resembling a light source or a reaction site. Thin, faint lines in a light yellow-green color crisscross the background.

7A GASES

UNIT 02: HOW REACTIONS HAPPEN

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 7

1. Define gas pressure and its units.
2. Calculate the number of moles, pressure, volume, or temperature in a sample of gas using the quantities of the other three.
3. Calculate the properties of each gas in a mixture of gases.
4. Predict how the pressure in a fixed-volume container will change after a complete reaction.
5. Use the kinetic molecular theory to explain the behavior of gases.
6. Describe the relationship between molecular mass and speed.
7. Calculate the relative speeds and molecular masses of gases using root mean square speed and Graham's law of effusion.

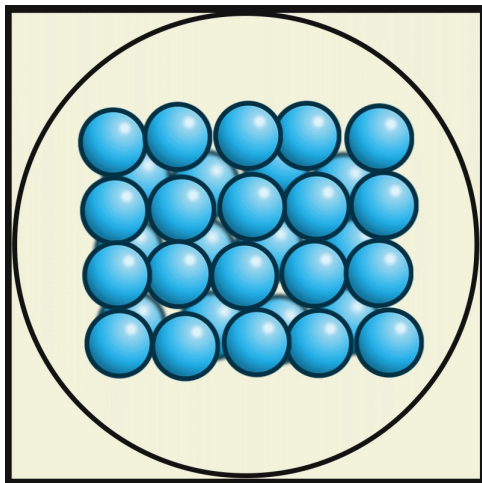
Gas Pressure

7.1

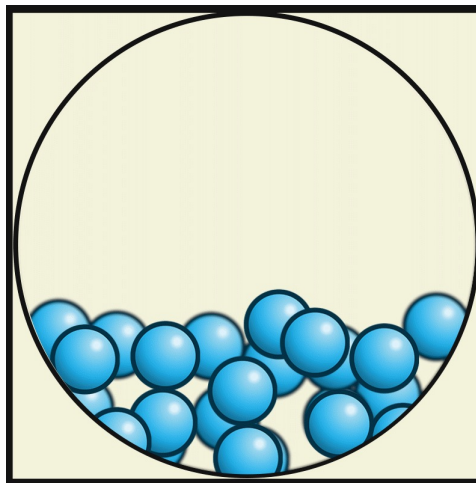
Matter occurs in three states, or phases: solids, liquids, and gases.

Gases do not have confined intrinsic volumes:

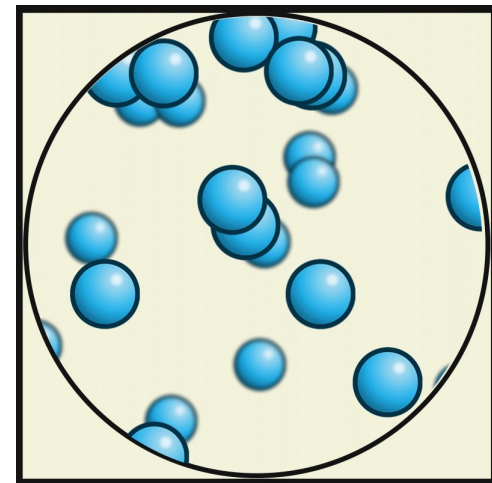
- They expand to fill the volume of the container.
- In other words: volume of gas is volume of container



SOLID



LIQUID



GASES

Gas Pressure

7.1

Pressure: Force exerted per unit area $P = \frac{F}{A}$

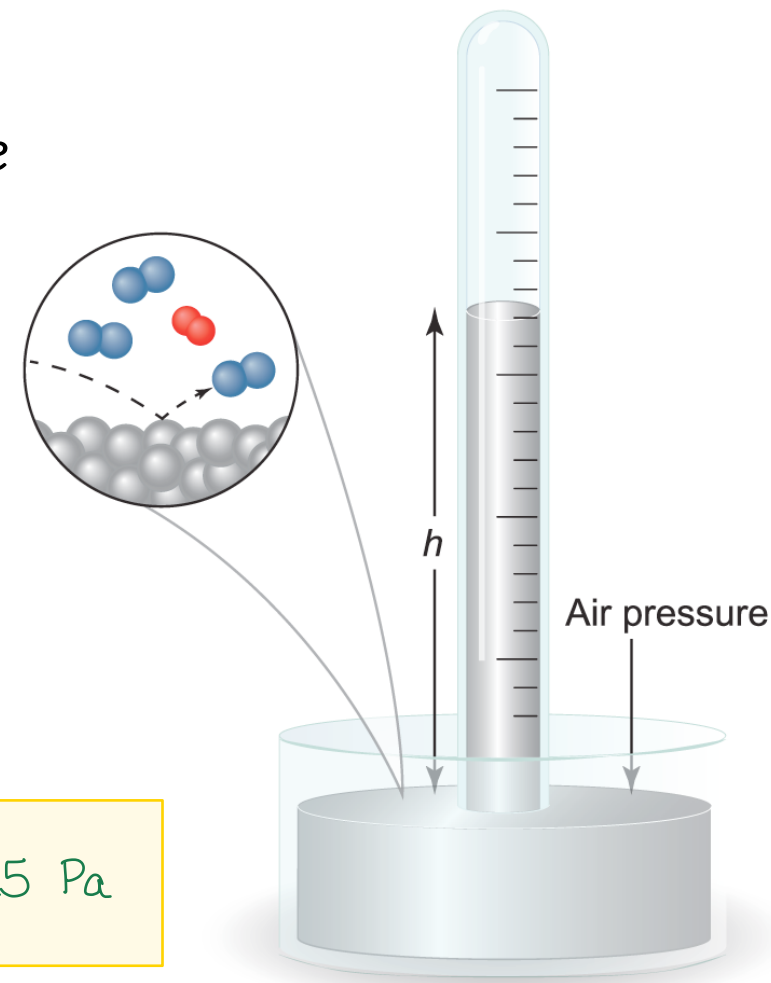
Units of pressure: Because pressure is an important *measurable* property of a gas, several different ways and units have been devised to measure it.

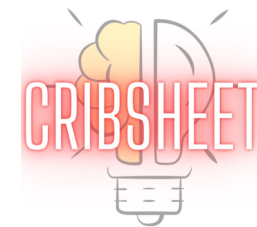
- Atmospheres (atm)
- Millimeters of mercury (mmHg or Torr)
- Pascal and kilopascal (Pa and kPa)

SI unit of pressure

$$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mmHg} = 101.325 \text{ kPa} = 101325 \text{ Pa}$$

Figure 7.3 Simple Barometer





Gas Pressure

7.1

Standard temperature and pressure (STP)

- A specific set of temperature and pressure conditions for gas.
- These are somewhat *arbitrary* definitions that allow us (scientists) to make “fair” *comparison* between gases.
- For a gas, **STP** is defined as: 1 atm and $0^\circ \text{C} = 273.15 \text{ K}$

Gas Pressure

7.1

For gases, we need to consider:

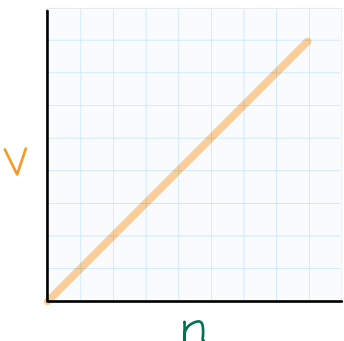
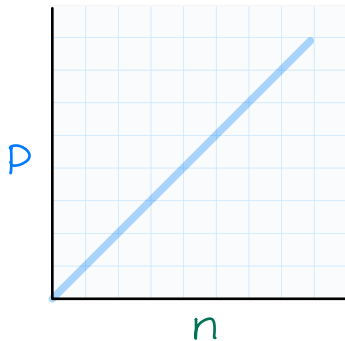
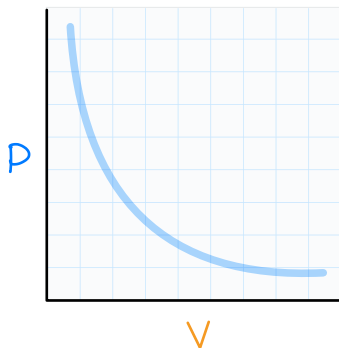
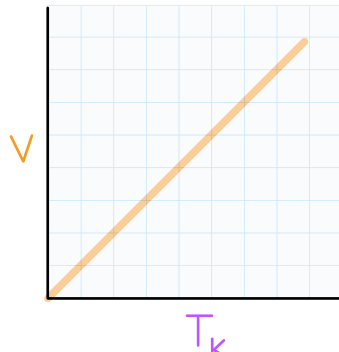
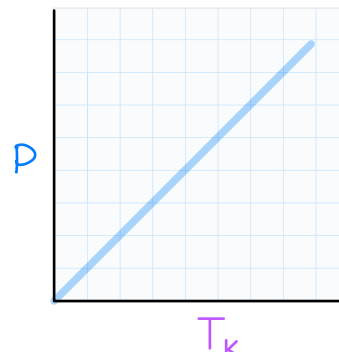
- Pressure (P)
 - Volume (V)
 - Temperature (T)
 - Moles (n)
-
- All of these properties are important for gases.
 - They are not *as important* when talking about solids and/or liquids.

Let's look at the relationships between these four properties for gases.

Empirical Gas Laws (Relationships)

7.2, 7.3, 7.5

- To observe relationships (gas laws), you need to hold the other properties constant.
- Temperature must be measured in the absolute Kelvin (K) scale: $T_K = T_{\circ C} + 273.15$

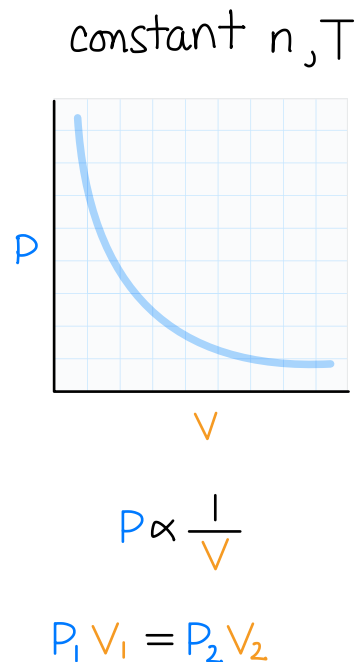
V vs. n	P vs. n	P vs. V	V vs. T	P vs. T
(constant P, T)	(constant V, T)	(constant n, T)	(constant n, P)	(constant n, V)
				
$V \propto n$	$P \propto n$	$P \propto \frac{1}{V}$	$V \propto T_K$	$P \propto T_K$
$\frac{V_1}{n_1} = \frac{V_2}{n_2}$	$\frac{P_1}{n_1} = \frac{P_2}{n_2}$	$P_1 V_1 = P_2 V_2$	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$

EXAMPLE PROBLEM

Learning Objective(s): 7.2

A gas sample has an initial pressure of 547 mmHg and an initial volume of 0.500 L.

What is the final pressure (in atm) when the volume of the gas is decreased to 225 mL?
Assume a constant temperature and constant number of moles of gas.



$$P_1 = 547 \text{ mmHg} \times \frac{1 \text{ atm}}{760 \text{ mmHg}} = 0.7197 \text{ atm} \longrightarrow P_2 = ?$$

$V_1 = 0.500 \text{ L}$ $V_2 = 0.225 \text{ L}$

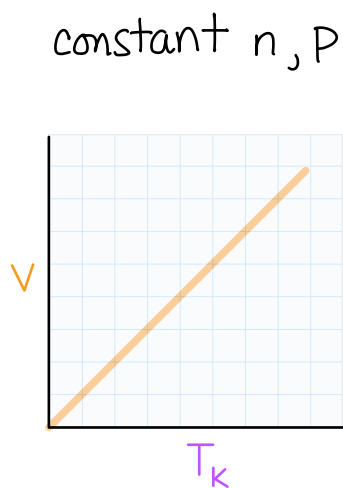
$$P_1 V_1 = P_2 V_2$$
$$(0.7197 \text{ atm})(0.500 \text{ L}) = P_2 (0.225 \text{ L})$$
$$P_2 = (0.7197 \text{ atm}) \cdot \frac{0.500 \text{ L}}{0.225 \text{ L}}$$

$P_2 = 1.60 \text{ atm}$ increases

IN-CLASS QUESTION 1

Learning Objective(s): 7.2

By what **ratio/factor** does the volume of a gas increase when its temperature is increased from **9.0 °C to 27.0 °C** at **constant pressure**?



$$V \propto T_K$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$V_2 = V_1 \cdot \underbrace{\frac{T_2}{T_1}}_{\text{ratio}}$$

temperature must be in Kelvin!

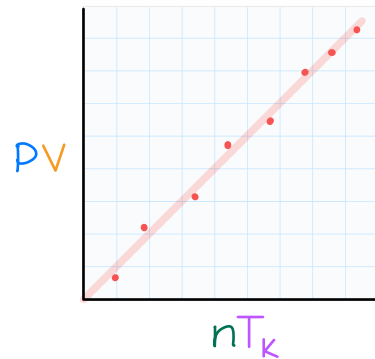
$$T_1 = 9.00^\circ\text{C} + 273.15 = 282.15\text{ K}$$
$$T_2 = 27.00^\circ\text{C} + 273.15 = 300.15\text{ K}$$
$$\frac{T_2}{T_1} = \frac{300.15\text{ K}}{282.15\text{ K}} = 1.0638 \text{ ; increases}$$

Ideal Gas Law

7.6

The relationships between P , V , n , and T are empirical (based on observations).

- Put together, they lead to the ideal gas law.



for any ideal gas!

$$\frac{PV}{nT} = \text{constant} = R$$

$$PV = nRT$$

pressure (atm) — P

volume (L) — V

moles (mol) — n

temperature (K) — T

universal gas constant — R

$$R = 0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}$$

The universal gas constant, R

- R has several possible sets of units. It is easiest to always use one “version” of R , and then match the units of P , V , n , and T with R .

IN-CLASS QUESTION 2

Learning Objective(s): 7.2

A 5.43 g sample of an elemental diatomic gas occupies 4.38 L at 25°C and a pressure of 0.950 atm . What is the identity of the gas?

General Approach: Find molar mass (mm) to identify element X_2 !

Use ideal gas law to get moles (n). Determine molar mass of X_2 .

$$PV = nRT$$

$$(0.950 \text{ atm})(4.38 \text{ L}) = n \left(0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \right) (25.0^\circ\text{C} + 273.15)$$

$$n = \frac{(0.950 \text{ atm})(4.38 \text{ L})}{\left(0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \right) (298.15 \text{ K})}$$

$$n = 0.170, \text{ mol } X_2$$

$$R = 0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}$$

$$\text{mm} = \frac{\text{mass (g)}}{\text{moles (n)}}$$

$$= \frac{5.43 \text{ g}}{0.170, \text{ mol}}$$

$$\text{mm} = 31.9 \text{ g/mol } X_2$$

Identify X_2 from PT. $\leftarrow$

$$X_2 = \text{O}_2, \text{ oxygen}$$

The Combined Gas Law

7.4

A relationship derived from a combination of the other gas laws to demonstrate the relationship between the pressure, volume, and temperature of a sample of gas when only n is constant.

changing constants

$$PV = nRT$$

$$\frac{PV}{T} = nR$$

Some #

initial

$$\begin{array}{c} P_1 \\ V_1 \\ T_1 \end{array} \quad n$$

final

$$\begin{array}{c} P_2 \\ V_2 \\ T_2 \end{array} \quad n$$

$$\frac{P_1 V_1}{T_1} = nR = \frac{P_2 V_2}{T_2}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

EXAMPLE PROBLEM

Learning Objective(s): 7.2

You have a gas initially at a pressure of 12 atm, a volume of 23 L, and a temperature of 200.0 K. You then raise the pressure to 14 atm and heat it to 300.0 K. **What is the new volume (in L) of the gas?**

$$\begin{array}{ll} P_1 = 12 \text{ atm} & P_2 = 14 \text{ atm} \\ V_1 = 23 \text{ L} & \Rightarrow V_2 = ? \\ T_1 = 200.0 \text{ K} & T_2 = 300.0 \text{ K} \end{array}$$

$$\text{constant } n : \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$V_2 = \frac{P_1}{P_2} \cdot \frac{T_2}{T_1} \cdot V_1$$

$$= \frac{12 \text{ atm}}{14 \text{ atm}} \cdot \frac{300.0 \text{ K}}{200.0 \text{ K}} \cdot (23 \text{ L})$$

$$V_2 = 30. \text{ L}$$

Dalton's Law of Partial Pressures

7.7

Dalton's law of partial pressures

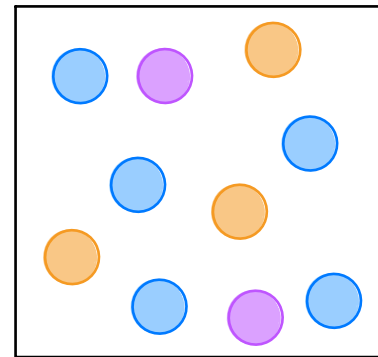
- The total pressure of a mixture of gases (P_{tot}) is the sum of the **partial pressures** of the component gases (P_i) in a mixture.
- We calculate the partial pressure (P_i) from the ideal gas law by assuming that each gas component (i) acts independently and occupies the entire volume.

$$P_{\text{tot}} = P_1 + P_2 + \dots + P_n = \sum P_i$$

$$P_i = \frac{n_i RT}{V}$$

$$P_{\text{tot}} = \frac{n_{\text{tot}} RT}{V}$$

5 mol Blue
3 mol Orange
2 mol Purple



all same V, T

$$P_{\text{tot}} = P_B + P_O + P_P$$

$$P_B = \frac{n_B RT}{V}$$

$$P_O = \frac{n_O RT}{V}$$

$$P_P = \frac{n_P RT}{V}$$

Dalton's Law of Partial Pressures

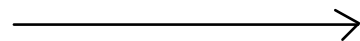
7.7

Dalton's law of partial pressures

- We can also calculate partial pressures of individual gases (P_i) based on the **mole fraction** of each gas component (χ_i).

$$\frac{P_i}{P_{\text{tot}}} = \frac{\frac{n_i RT}{V}}{\frac{n_{\text{tot}} RT}{V}} = \frac{n_i}{n_{\text{tot}}} = \chi_i$$

mole fraction



$$P_i = \chi_i P_{\text{tot}}$$

another way to calculate P_i
if you know χ_i , P_{tot}

EXAMPLE PROBLEM

Learning Objective(s): 7.3

A gaseous mixture contains 15.50 mol of total gases. If the mixture is known to contain 4.50 mol O_2 with a partial pressure of 3.10 atm , what is the total pressure (in atm) of the system?

mole fraction: $\chi_i = \frac{n_i}{n_{\text{tot}}}$

$$\begin{aligned}\chi_{\text{O}_2} &= \frac{n_{\text{O}_2}}{n_{\text{tot}}} \\ &= \frac{4.50 \text{ mol O}_2}{15.50 \text{ mol total}}\end{aligned}$$

$$\chi_{\text{O}_2} = 0.29$$

partial pressure: $P_i = \chi_i P_{\text{tot}}$

$$P_{\text{O}_2} = \chi_{\text{O}_2} \cdot P_{\text{tot}}$$

$$3.10 \text{ atm} = (0.29) \cdot P_{\text{tot}}$$

$$P_{\text{tot}} = 11 \text{ atm}$$

IN-CLASS QUESTION 3

Learning Objective(s): 7.3

$$mm = 28.01 \text{ g/mol}$$

$$mm = 44.009 \text{ g/mol}$$

5.00 g of CO and 5.00 g of CO₂ are placed in a 0.7500 L container at 323.15 K. What is the total pressure (in atm) in the container?

Method 1: From partial pressures

$$n_{\text{CO}} = 5.00 \text{ g CO} \times \frac{1 \text{ mol CO}}{28.01 \text{ g CO}} = 0.1785 \text{ mol CO}$$

$$n_{\text{CO}_2} = 5.00 \text{ g CO}_2 \times \frac{1 \text{ mol CO}_2}{44.009 \text{ g CO}_2} = 0.1136 \text{ mol CO}_2$$

$$P_{\text{CO}_2} = \frac{n_{\text{CO}_2} R T}{V} = \frac{(0.1136 \text{ mol}) \left(0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right) (323.15 \text{ K})}{0.7500 \text{ L}} = 4.017 \text{ atm}$$

$$P_{\text{CO}} = \frac{n_{\text{CO}} R T}{V} = \frac{(0.1785 \text{ mol}) \left(0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right) (323.15 \text{ K})}{0.7500 \text{ L}} = 6.311 \text{ atm}$$

$$P_{\text{tot}} = P_{\text{CO}} + P_{\text{CO}_2} = 10.33 \text{ atm}$$

Method 2: From total moles

$$n_{\text{tot}} = n_{\text{CO}} + n_{\text{CO}_2} = 0.2921 \text{ mol total gas}$$

$$P_{\text{tot}} = \frac{n_{\text{tot}} R T}{V} = \frac{(0.2921 \text{ mol}) \left(0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right) (323.15 \text{ K})}{0.7500 \text{ L}}$$

$$P_{\text{tot}} = 10.3 \text{ atm}$$

BONUS QUESTION

Assessed on: Fall 2021, Exam 2

A gaseous mixture contains 1.06 mol of CO_2 , 1.14 mol Ar, and 1.32 mol of Xe. If the partial pressure of Ar is 0.793 atm, what is the partial pressure (in atm) of the CO_2 ?